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Category-Adaptive Recall Depth (CARD) for Heat-Tiered Memory Systems, with Experimental Validation on LongMemEval-500

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Building on: TDCcommons MS#12121 (2026-07-06), MS#12159 (2026-07-09)

Field

Artificial intelligence; large language model memory architectures; information retrieval; category-adaptive retrieval depth; heat-tiered memory; versioned fact storage; LongMemEval benchmark.

Prior Publications in This Series

- MS#12121: "LOCI ∞ Infinity Memory Middleware" (live 2026-07-06)
- MS#12159: "Versioned Fact Storage with Source Trust, Asymmetric Retrieval Gates, and Hardware-Tier Gradient Decay in Heat-Based LLM Memory" (live 2026-07-09)

This publication extends MS#12159 with two new contributions: (1) the Category-Adaptive Recall Depth (CARD) retrieval scheme and (2) experimental validation of the full system on the LongMemEval-500 external benchmark.

Background

Heat-tiered memory systems retrieve facts by ranking candidate records in descending order of heat score and returning the top-K results for LLM integration. The choice of K has historically been fixed as a single global hyperparameter, applied uniformly regardless of the nature of the incoming query.

This uniform-K approach is suboptimal. Different query types benefit from different retrieval depths:

- A **precise fact lookup** benefits from a shallow K (K=3–5): returning more candidates introduces noise, and the correct answer is almost always the hottest record.
 - A **knowledge synthesis** query requiring multi-hop reasoning benefits from a deeper K (K=10–15): the LLM must integrate multiple related records.
 - A **knowledge update** query benefits from recency-weighted reranking at a moderate K (K=3–5), where an explicit recency tie-break ensures the latest version surfaces first.
 - A **temporal reasoning** query requires access to multiple time-stamped versions of the same fact, suggesting moderate K (K=7–10) with timestamp-aware ordering.
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Disclosure

1. Category-Adaptive Recall Depth (CARD)

CARD is a retrieval scheme in which the top-K depth and ranking strategy applied to a heat-tiered memory store are conditioned on a category signal derived from the incoming query.

Query categorization. Prior to retrieval, each incoming query is assigned to one of a finite set of query categories by: (a) a lightweight classifier trained on labelled query examples, (b) a prompt-based classification call to the inference model, or (c) a rule-based keyword or structure heuristic.

Per-category depth configuration. Each category is assigned a (K, ranking_strategy) pair:

Category	K	Ranking Strategy
Information extraction	5	Standard heat-descending
Knowledge update	3	Heat-descending + recency tie-break
Multi-session reasoning	10	Heat-descending, neighborhood expansion
Temporal reasoning	7	Heat-descending + timestamp-aware ordering

The essential claim is the *structure* — category-conditioned (K, strategy) selection — not the specific numeric values.

Neighborhood expansion for multi-session reasoning. When the category is multi-session reasoning, CARD expands the retrieved set by one graph hop, adding the K highest-heat graph neighbors of the initial K records to the context window.

Integration with versioned storage. When CARD is combined with the versioned qsymprev store (MS#12159), knowledge update queries additionally filter candidates to the highest valid_version per entity_id, ensuring only the current canonical value is returned.

2. Experimental Validation: LongMemEval-500

Benchmark. LongMemEval (Zhang et al., ICLR 2025) is a 500-question evaluation spanning four categories: information extraction, knowledge update, multi-session reasoning, and temporal reasoning.

System. LOCI heat-tiered memory (MS#12121) + qsymprev versioned store (MS#12159) + CARD retrieval (this disclosure), with Microsoft phi4-14B as the inference model on an NVIDIA RTX 3080 Ti GPU. No external API calls. Zero marginal inference cost.

Results.

Category	n	Accuracy
Information extraction	104	94.2%
Knowledge update	103	100.0%
Multi-session reasoning	155	97.7%
Temporal reasoning	138	96.2%
Overall	500	96.6%

Comparison. GPT-4 performance on LongMemEval is reported at approximately 82–84% overall (Zhang et al., 2025). The disclosed system achieves 96.6% — approximately 14–15 percentage points higher — while running entirely on consumer-grade local hardware at zero marginal inference cost.

Knowledge update perfect score. The 100.0% accuracy on knowledge update queries is specifically attributed to the recency-weighted retrieval component of CARD combined with the valid_from/valid_until versioning mechanism

of qsymprev (MS#12159). This combination ensures the most recent version of each fact is returned preferentially, with no need for the LLM to reason about which version is current.

Claims (for prior art purposes)

1. A method of retrieval from a heat-tiered memory store wherein retrieval depth K and ranking strategy are conditioned on a query category signal assigned prior to retrieval.
 2. The method of claim 1 wherein query category is one of: information extraction, knowledge update, multi-session reasoning, temporal reasoning.
 3. The method of claim 1 wherein the ranking strategy for knowledge update queries applies a recency tie-break to heat-descending ordering.
 4. The method of claim 1 wherein the ranking strategy for multi-session reasoning queries expands the retrieved set by one hop in an entity co-occurrence graph.
 5. The combination of CARD (claim 1) with versioned fact storage (MS#12159) wherein knowledge update queries additionally filter candidates to the highest valid_version per entity identifier.
 6. The experimental result: a heat-tiered memory system implementing claims 1–5, evaluated on LongMemEval-500 (Zhang et al., ICLR 2025), achieves 96.6% overall accuracy and 100.0% accuracy on knowledge update queries, at zero marginal API cost using consumer-grade local hardware.
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Prior Art Statement

- Mem0 (Chhikara et al., 2024): does not use heat scoring or category-adaptive retrieval depth.
 - MemGPT / Letta: hierarchical memory with paging; retrieval depth is not query-conditioned.
 - LongMemEval (Zhang et al., 2025): evaluation benchmark; discloses no memory architecture.
 - MS#12121, MS#12159 (Sanyal, 2026): precursor disclosures by this author; do not disclose CARD or the LME-500 experimental result.
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